Stochastic Modeling and Simulation
3rd Y - Semester 5
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Answer Key - Test

Exercise 1 *Let H:"The player wins 10 DA" and T: "the player loses 10 DA"*
We have: $p = \mathbb{P}(H) = \frac{3}{4}$ *and* $\mathbb{P}(T) = \frac{1}{4}$
We use the inverse transform method for a discrete r.v. to simulate the game
Algorithm

1. *For* $i = 1, n$ *and* $X_0 = 0$
2. *Generate* $U \sim \mathcal{U}[0,1]$
3. *If* $u \leq p$, *then* $X = 10$
   else $X = -10$ (1,5)
4. $X = X_0 + X$

Application: *For* $p = 0.75$ *and* $p = 0.5$

u_i	$p = 0.75$	*Win*	$p = 0.5$	*Win*
0.404190	H	$+10$	H	$+10$
0.735378	H	$+10$	T	-10
0.369251	H	$+10$	H	$+10$
0.705100	H	$+10$	T	-10
0.833575	T	-10	T	-10
0.397475	H	$+10$	H	$+10$
0.151053	H	$+10$	H	$+10$
0.925697	T	-10	T	-10
0.928252	T	-10	T	-10
0.481475	H	$+10$	H	$+10$
$\sum$	-	$+40$	$-$	0

(2,5)

Exercise 2 *Consider the triangular random variable X with pdf*

$$f(x) = \begin{cases} 0 & \text{if} \quad x < 2a \text{ or } x \geq 2b \\ \frac{x-2a}{(b-a)^2} & \text{if} \quad 2a \leq x < a+b \\ \frac{2b-x}{(b-a)^2} & \text{if} \quad a+b \leq x < 2b \end{cases}$$

1. *Derive the corresponding cdf:* $F_X(x) = \int\limits_{-\infty}^{x} f(t)\,dt$

   If $x < 2a : F_X(x) = \int\limits_{2a}^{x} 0 dt = 0$

   If $2a \leq x < a+b : F_X(x) = \int\limits_{2a}^{x} \frac{t-2a}{(b-a)^2} dt = \frac{(x-2a)^2}{2(b-a)^2}$

   If $a+b \leq x < 2b : F_X(x) = \int\limits_{2a}^{a+b} \frac{t-2a}{(b-a)^2} dt + \int\limits_{a+b}^{x} \frac{2b-t}{(b-a)^2} dt = \frac{1}{2} - \frac{(2b-x)^2}{2(b-a)^2} + \frac{(b-a)^2}{2(b-a)^2}$

So

$$F_X(x) = \begin{cases} 0 & \textit{if} \quad x < 2a \\ \frac{(x-2a)^2}{2(b-a)^2} & \textit{if} \quad 2a \le x < a+b \\ 1 - \frac{(2b-x)^2}{2(b-a)^2} & \textit{if} \ a+b \le x < 2b \\ 1 & \textit{if} \quad x \ge 2b \end{cases} \qquad (3.0)$$

2. *Simulation of* X *using the inverse transform method: Let* $U \sim \mathcal{U}[0,1]$

   If $0 \le U < \frac{1}{2} : u = \frac{(x-2a)^2}{2(b-a)^2} \Rightarrow x = 2a + (b-a)\sqrt{2u}$.

   If $\frac{1}{2} \le U \le 1 : u = 1 - \frac{(2b-x)^2}{2(b-a)^2} \Rightarrow x = 2b + (b-a)\sqrt{2(1-u)}$.

   Algorithm

   1) Generate $U \sim \mathcal{U}[0,1]$

   2) If $U < \frac{1}{2} : X = 2a + (b-a)\sqrt{2u}$, (1,5)

   else: $X = 2b + (b-a)\sqrt{2(1-u)}$.

   Application*: for* $a = 1$ *and* $b = 2$*, generate three observations of this variable:*

   $u_1 = 0.404190 < \frac{1}{2}$, *so* $X = 2 + \sqrt{2u} = 2.8991$

   $u_1 = 0.735378 > \frac{1}{2}$, *so* $X = 4 - \sqrt{2(1-u)} = 3.2726$ (1,5)

   $u_1 = 0.369251 < \frac{1}{2}$, *so* $X = 2 + \sqrt{2u} = 2.85936$